

Week 1: Topological Spaces and Continuous Functions

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Review: topology of $\mathbb{R}$

Let A, B be two sets.

We say that $|A| \leq |B|$ if there exists an injection $f : A \rightarrow B$.

We say $|A| = |B|$ if there exists a bijection between A and B .

It is useful to have the following theorem of Schröder-Bernstein:

$|A| = |B|$ if and only if $|A| \leq |B|$ and $|B| \leq |A|$.

A set A is countable if $|A| \leq |\mathbb{N}|$. A set A is infinite if $|\mathbb{N}| \leq |A|$.

Fact: $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$, which is strictly bigger than $|\mathbb{N}|$.

A set $U \subset \mathbb{R}$ is called open if for any $x \in U$, there is an open interval contained in U that contains x , i.e. $(x - \delta, x + \delta) \subset U$ with $\delta > 0$.

Notice that $\emptyset$ and $\mathbb{R}$ are open. An open set is necessarily a union of open intervals, as we can take the union of each $(x - \delta, x + \delta)$ above for all $x \in U$.

A set $U \subset \mathbb{R}$ is a neighborhood of $x \in \mathbb{R}$, if there is an open interval contained in U that contains x .

Therefore, a set $U \subset \mathbb{R}$ is open if and only if it is a neighborhood for all $x \in U$.

Finally a set $F \subset \mathbb{R}$ is closed if and only if its complement F^c is open in $\mathbb{R}$.

An arbitrary union of open sets in $\mathbb{R}$ is open.

A finite intersection of open sets in $\mathbb{R}$ is open.

But $\bigcap_{n \geq 0} (-1/n, 1/n) = \{0\}$, which is not open.

Example: the set of rational numbers $\mathbb{Q} \subset \mathbb{R}$ is not open, nor closed.

A sequence of real numbers x_n converges to some $x \in \mathbb{R}$ if and only if for every neighborhood V_x of x , for all sufficiently large n (i.e. $\exists N > 0, \forall n \geq N$), $x_n \in V_x$.

A set $F \subset \mathbb{R}$ is closed if and only if every limit of every convergent sequence in F belongs to F .

Let $A \subset \mathbb{R}$ be a subset of $\mathbb{R}$. Then $x \in A$ is called

1. an interior point of A if there exists $\delta > 0$ such that $(x - \delta, x + \delta) \subset A$;
2. an isolated point of A if there exists $\delta > 0$ such that x is the only point of A belonging to $(x - \delta, x + \delta)$;
3. a boundary point of A if for every $\delta > 0$, the interval $(x - \delta, x + \delta)$ intersects both A and A^c ;
4. an accumulation point of A if for every $\delta > 0$, the interval $(x - \delta, x + \delta)$ contains some other point of A other than x .

Important: rephrase these definitions in terms of **neighborhoods**.

A set of real numbers is connected if and only if it is an interval.

Therefore, a connected set in $\mathbb{R}$ is the same as path connected, or convex.

However the **true definition** (with only open sets) is that a connected set A cannot be written as the disjoint sum of two non-empty open sets in A .

A set of real numbers is compact if and only if it is closed and bounded (Heine-Borel). Or if and only if every sequence has a convergent subsequence (Bolzano-Weierstrass).

However the **true definition** (with only open sets) is that a set is compact if and only if every open cover has a finite subcover.

Example: a new definition of
continuity

Consider a map between sets $f : E \rightarrow F$. For any subset $Y \subset F$, the **preimage** of Y by the map f is defined as

$$f^{-1}(Y) = \{x \in E ; f(x) \in Y\}.$$

It is a subset of E and this is well-defined for any map.

Notice that for the direct image $f(\cup A_i) = \cup f(A_i)$, $f(\cap A_i) \subset \cap f(A_i)$, but for the preimage, $f^{-1}(\cup B_i) = \cup f^{-1}(B_i)$, $f^{-1}(\cap B_i) = \cap f^{-1}(B_i)$.

The $\epsilon - \delta$ definition of the continuity revisited

Recall the $\epsilon - \delta$ definition for a function $f : \mathbb{R} \rightarrow \mathbb{R}$. The **continuity** of the function f at a point a with $y = f(a)$ is defined as

$$\forall \epsilon > 0, \exists \delta > 0, \forall |x - a| < \delta, \quad |f(x) - f(a)| < \epsilon.$$

Let us rewrite this in terms of **preimage**:

$$\forall \epsilon > 0, \exists \delta > 0, (a - \delta, a + \delta) \subset f^{-1}((f(a) - \epsilon, f(a) + \epsilon)).$$

That is, the preimage of any **neighborhood** of $y = f(a)$ is a **neighborhood** of a .

In fact, the continuity above is equivalent to the very general statement:

The preimage of any open set by f is an open set.

We only need to know about **open sets** to define continuity.

Theory: topological spaces and
continuous functions

A **topological space** is a set X equipped with a collection $\mathcal{O}$ of **open sets** $O \subset X$. That is, a set $O \subset X$ is called open if and only if $O \in \mathcal{O}$. We require that:

1. Trivial sets are open: $\emptyset, X \in \mathcal{O}$;
2. Closure under **finite intersection**: if $U, V \in \mathcal{O}$ then $U \cap V \in \mathcal{O}$;
3. Closure under **arbitrary union**: if $\mathcal{A} \subset \mathcal{O}$ then $\bigcup_{O \in \mathcal{A}} O \in \mathcal{O}$.

Informally speaking, finite intersection of open sets is open and any union of open sets is open.

A **topology** on X is simply the declaration of open sets $(X, \mathcal{O})$.

A set F is called **closed** (“fermé” in French) if its complement is open, i.e. if $X - F$ is open or $X - F \in \mathcal{O}$. You should check that

1. Trivial sets are closed;
2. Closure under **finite union**;
3. Closure under **arbitrary intersection**.

Equivalently, you can specify a topology by giving all the closed sets, but the convention is to use open sets.

Example: two basic topologies

Given a set X , there are always two possible topologies:

1. **Trivial topology**: the topology where only $\emptyset$ and X are open;
2. **Discrete topology**: the topology where every subset of X is open.

Two topologies are the **same** if all the open sets are the same, $\mathcal{O} = \mathcal{O}'$.

A topology $\mathcal{O}'$ is **finer** than $\mathcal{O}$ if $\mathcal{O} \subset \mathcal{O}'$. We also say that $\mathcal{O}$ is **coarser** than $\mathcal{O}'$.

Two topologies might not be necessarily **comparable**.

Then the finest topology is the discrete topology, and the least fine (=smallest) topology is the trivial topology (also called **indiscrete topology**).

We often omit $\mathcal{O}$ and write only X instead of $(X, \mathcal{O})$ for a topological space.

Basis of a topology

For $\mathbb{R}$, it is rather hard to define a generic open set. Instead, we implicitly used the following definition (you should verify that this defines indeed a topology):

A **basis** $\mathcal{B}$ of a topology is a collection of sets such that

1. $X = \bigcup_{B \in \mathcal{B}} B$;
2. If $x \in X$, $B_1, B_2 \in \mathcal{B}$ such that $x \in B_1$ and $x \in B_2$, then there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subset B_1 \cap B_2$.

Then we say that a set O is open if it is a **union of elements** in $\mathcal{B}$ (called **generated** by $\mathcal{B}$, i.e. for all $x \in O$, there is some $B \in \mathcal{B}$ such that $x \in B \subset O$).

For example, a basis for the usual topology on $\mathbb{R}^n$ is given by the collection of open balls $B(x, \epsilon)$ for $x \in \mathbb{R}^n$ and $\epsilon > 0$.

In general, the topology $\mathcal{O}$ **generated** by a list of properties is the **smallest topology** for which these properties hold.

Equivalently, a family $\mathcal{B}$ of subsets of X is a basis for the topology $\mathcal{O}$ iff

1. Every element in $\mathcal{B}$ is in $\mathcal{O}$;
2. For every open set O in $\mathcal{O}$ and point $x \in O$, there is some basic open set $B \in \mathcal{B}$ such that $x \in B \subset O$.

Proof.

$\implies$ Every element in $\mathcal{B}$ can be written as a union with one element (itself): thus the first point. For the second point, if $x \in O$ and O is union of elements in $\mathcal{B}$, then $x \in B$ for some $B \in \mathcal{B}$.

$\impliedby$ Take $O \in \mathcal{O}$, $x \in O$ and denote by $B_x \in \mathcal{B}$ such that $x \in B_x \subset O$ by the second point. Then O is the union of all such B_x , and this also applies to the open set $O = X$. Now take $B_1, B_2 \in \mathcal{B} \subset \mathcal{O}$: then $B_1 \cap B_2$ is open and the second point yields some $B_3 \in \mathcal{B}$ such that $x \in B_3 \subset B_1 \cap B_2$. $\square$

We have the following generalized definition of continuity:

A function $f : X \rightarrow Y$ between two topological spaces is called **continuous** if $f^{-1}(U)$ is open in X for any open set $U \subset Y$.

That is, the **preimage** by a continuous function of an open set is an open set.

Intuitively, this says “ $f(x)$ is close to $f(a)$ whenever x is close to a ”.

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1. Composition of continuous functions is continuous.
 2. It suffices to verify the preimage condition on a basis $\mathcal{B}$ of Y .

For example, to check that $f : X \rightarrow \mathbb{R}$ is continuous, it suffices to show that $f^{-1}(a, b)$ is open for all $(a, b) \in \mathbb{R}$.

Homeomorphic spaces are topologically the same.

A bijection between spaces, $f : X \rightarrow Y$ is a **homeomorphism** if and only if f and f^{-1} are continuous.

In other words, O is open if and only if $f(O)$ is open.

A bijective continuous map is not always a homeomorphism. Example:

$$f : \mathbb{N} \rightarrow \{0\} \cup \{1/1, 1/2, 1/3, \dots\}, \quad f(0) = 0, \quad f(n) = 1/n, n \geq 1.$$

However, bijective continuous map between compact Hausdorff spaces (defined later in this course) is always a homeomorphism.

Example: $\mathbb{R}$ and $(0, 1)$

The overall size of a space is not something topologically invariant.

The real numbers $\mathbb{R}$ are homeomorphic to the open interval $(0, 1)$.

Hint: consider $\tan : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$.

One of the main goals of topology is to classify **topological invariants**, e.g. objects that are stable under homeomorphisms.

We can already define a lot of notions with only open and closed sets.

1. **Closure.** The closure $\overline{A}$ of $A \subset X$ is the smallest closed set containing A .
2. **Interior.** The interior $\mathring{A}$ of $A \subset X$ is the largest open set contained in A .
3. **Boundary.** The boundary ∂A of $A \subset X$ is the difference $\overline{A} - \mathring{A}$.
4. **Density.** We say $A \subset X$ is dense if and only if $\overline{A} = X$.

A point $x \in X$ lies on the **boundary** if and only if every neighborhood of x intersects both A and $X - A$. For example, the boundary of $\mathbb{Q}$ in $\mathbb{R}$ is $\mathbb{R}$.

A topological space with a **countable dense subset** is called **separable**.

Limit points and isolated points

We should generally replace the idea of convergence sequences by neighborhoods, since in general there is no metric to measure the distance to the (possible) limit.

We say that x is the **limit of a sequence** $\{x_n\}_{n \geq 0}$ if every neighborhood of x meets all but a finite subset of $\{x_n\}_{n \geq 0}$.

We say that x is a **limit point** of A if every neighborhood of x meets $A - \{x\}$.

Verify that in a metric space, this means that x is limit point if and only if there exists a sequence $a_n \in A$ such that $a_n \rightarrow x$.

Exercise: the closure $\overline{A}$ of A is the union of A and its limit points.

A point $x \in A$ is **isolated** if it has a neighborhood V such that $V \cap A = \{x\}$.

Exercise: every point of a closed set A is either a limit point or an isolated point.

The set of limit points A' of a closed set $A \subset \mathbb{R}$ is called the **derived set** of A , i.e. we throw away the isolated points.

The Cantor set K does not contain any interval yet $K' = K$. However, a countable closed set $A \subset [0, 1]$ always contain an isolated point (by Baire), so that $A' \neq A$.

On the other hand, a countable closed set $A \subset [0, 1]$ always has an isolated point, so that $A' \neq A$ (this we shall not prove). This led Cantor to the invention of ordinals and much of the set theory, but we will not discuss this in this course.

Example: sequences are not enough

Let $X = \mathbb{R}$ equipped with the topology $\mathcal{T}$ that closed sets are $\mathbb{R}$ and countable sets of $\mathbb{R}$. Take the ball $B = B(0, 1) \subset X$.

Then the closure of B is the smallest closed sets containing B , so $\overline{B} = \mathbb{R}$.

But there is no sequence in B that converges to $2 \in \overline{B}$, since every sequence in B is already closed!

Therefore, pay attention to the definition of **limit points**! We will see later that the sequential definition is very different from the neighborhood formalism.

We say X is **Hausdorff** if any pair of distinct points have disjoint neighborhoods.

It is important to notice that the limit of a sequence (if there is one) in a Hausdorff space is unique. This is not true e.g. for the trivial topology.

This is usually called the **T_2 axiom**. It implies the T_1 axiom that points are closed. For example, if X is infinite with the cofinite topology, then X is T_1 , but not T_2 .

The product of Hausdorff spaces is Hausdorff.

The subspace of a Hausdorff space is Hausdorff.

You should try to prove the above propositions.

Let $C(X, Y)$ be the set of all continuous functions $f : X \rightarrow Y$. We write $C(X) = C(X, \mathbb{R})$ for the set of continuous real-valued functions. Then $C(X)$ is an **algebra**, i.e. it is closed under the sum $f + g$ and the product fg , and contains 1.

In fact, if we take $Y = \{0, 1\}$ with a suitable topology, then the space $C(X, Y)$ uniquely determines the topology on X :

Consider Y with the topology that $\{1\}$ is open and $\{0\}$ is not. Then $A \subset X$ is open if and only if $\mathbf{1}_A$ is continuous.

A **neighborhood** of $x \in X$ is just an open set $V \subset X$ such that $x \in V$. Sometimes the set of neighborhood of x is denoted $\mathcal{V}(x)$.

We say that a sequence x_n **converges** to x if for any neighborhood V of x , $x_n \in V$ for all large enough n , i.e. $\forall V \in \mathcal{V}(x), \exists N \in \mathbb{N}, \forall n \geq N, x_n \in V$.

Consider the **cofinite topology**: $A \subset X$ is closed if and only if $|A| < \infty$ or $A = X$.

If x_n is a sequence in X , then $x_n \rightarrow x$ for any $x \in X$! There is **no uniqueness of limit** in this topology.

Example: the lower-limit topology

Consider the topology $\mathbb{R}_l$ on $\mathbb{R}$ generated by the intervals $[a, b)$, i.e. the latter form a basis of open sets. In this topology, $x_n \rightarrow x$ if and only if x_n converges to x in the usual sense and $x_n \geq x$ for all large enough n .

A function $f : \mathbb{R}_l \rightarrow \mathbb{R}$ is continuous if and only if $f(x_n) \rightarrow f(x)$ whenever $x_n \rightarrow x$ from above. The identity map $i : \mathbb{R}_l \rightarrow \mathbb{R}$ is continuous, but $i : \mathbb{R} \rightarrow \mathbb{R}_l$ is not.

The $\mathbb{R}_l$ is also called the **lower-limit or uphill topology**. It is easier to “go down from above” than to “climb up from below”. This lack of symmetry is not consistent with a metric: in fact $\mathbb{R}_l$ is not metrizable.

Example: the product topology

Let X and Y be topological spaces.

The **product topology** on $X \times Y$ is generated by sets of the form $U \times V$ with U open in X and V open in Y .

Notice that there are more open sets in $X \times Y$ than just those written as $U \times V$.

1. The product topology is the **smallest topology** such that the projections $X \times Y \rightarrow X$, $X \times Y \rightarrow Y$ are continuous.
2. A function $f : Z \rightarrow X \times Y$ such that $f(z) = (f_1(z), f_2(z))$ is continuous if and only if each of its coordinates f_1 and f_2 is continuous.

Example: with the usual $I = [0, 1] \subset \mathbb{R}$, the product topology on I^2 is the usual one.

Example: topology on the square

A set X with a **total ordering** can be made topological X_o by taking the intervals $(a, b) = \{x \in X ; a < x < b\}$ as a basis.

The **lexicographical ordered** square I_o^2 has a weirder topology. If $(a, b) \in I_o^2$ with $0 < b < 1$, then for any $b \in (b', b'') \subset (0, 1)$, the vertical segment $\{a\} \times (b', b'')$ is open in I_o^2 . So an open subset in the interior of I^2 in the usual sense is open for I_o^2 .

Things are harder on the boundary: consider a neighborhood of $(a, 1)$ in I_o^2 . If $0 < a < 1$, then it must contain some $\{a'\} \times S$, where S is a vertical segment. So a small neighborhood of $(a, 1)$ in I_o^2 is not open in the usual sense.

Indeed, in a dictionary, if $w = \text{Czech}$ is the last word in the section C , then any neighborhood of w contains some word beginning with D .

Example: the subspace topology

If $A \subset X$ and X is a topological space, the **subspace topology** on A is defined by

$$\mathcal{O}_A = \{U \cap A ; U \in \mathcal{O}_X\}.$$

This is the unique topology such that $f : Z \rightarrow A$ is continuous if and only if $\iota \circ f$ is continuous, where $\iota : A \rightarrow X$ is the **inclusion map**.

We will see later that this is the topology induced by the inclusion $\iota : A \rightarrow X$, i.e. the coarsest topology on A making ι continuous.

1. What is the subspace topology of $[0, 1] \subset \mathbb{R}$?
2. What is the subspace topology of $\mathbb{R} \cup i\mathbb{R} \subset \mathbb{C}$?
3. What is the subspace topology of the square I^2 as a subspace of the ordered plane $\mathbb{R}_o^2$? Answer: it is not the ordered square I_o^2 .